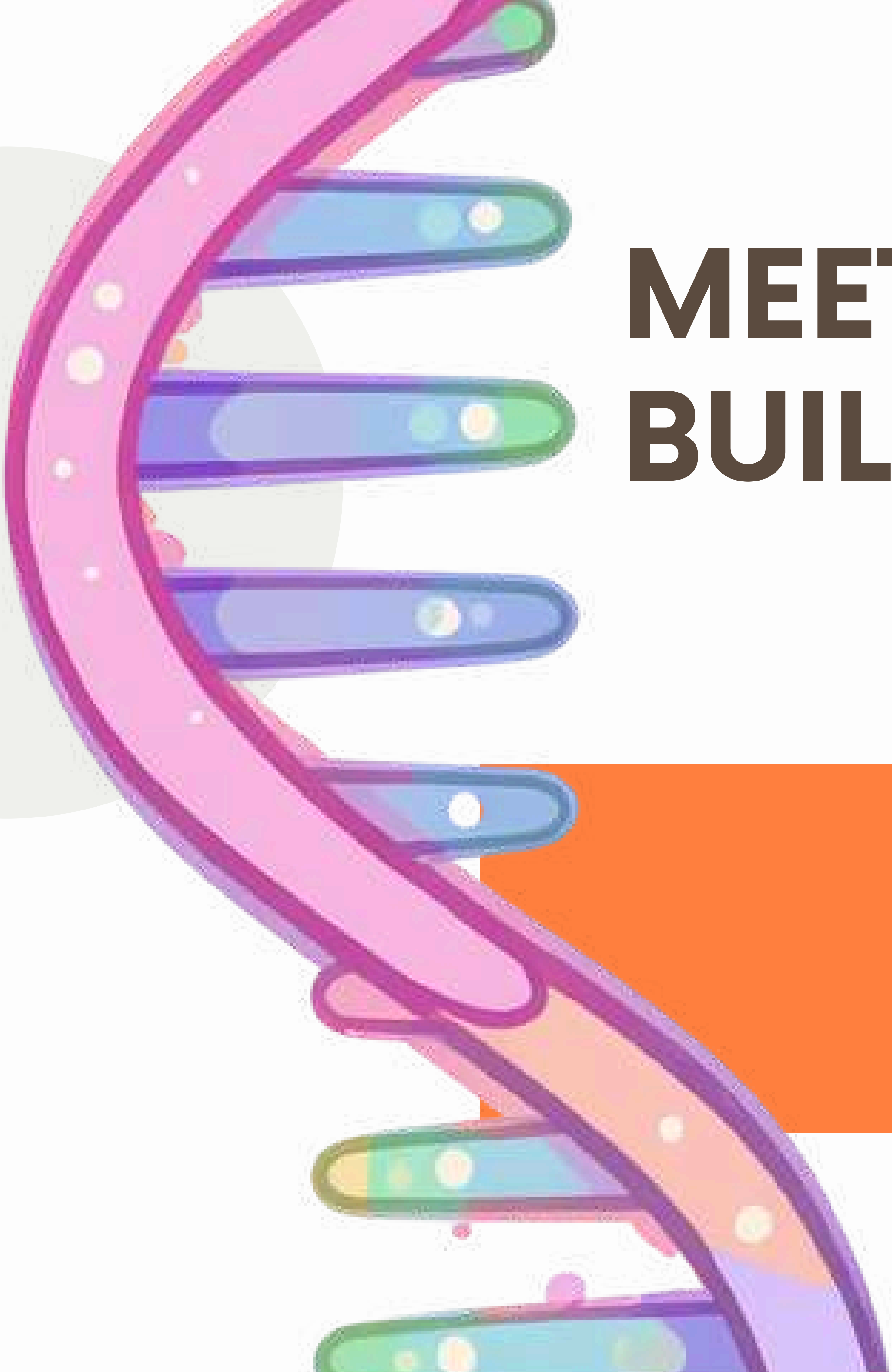
BIOTECHNOLOGY: DNA EXTRACTION

Discover the tiny world inside you!

WHAT IS BIOTECHNOLOGY?

Biotechnology means using living things to solve problems and make useful products!

Examples include making medicines to help sick people, improving crops for farmers, and cleaning up the environment. Scientists use tiny living things to create amazing solutions!



MEET DNA — THE BUILDING BLOCKS OF LIFE

DNA carries the instructions for how all living things grow and function. Think of it like a recipe book inside every cell that tells your body how to build and run itself!

WHY EXTRACT DNA?

Scientists extract DNA for many important reasons:

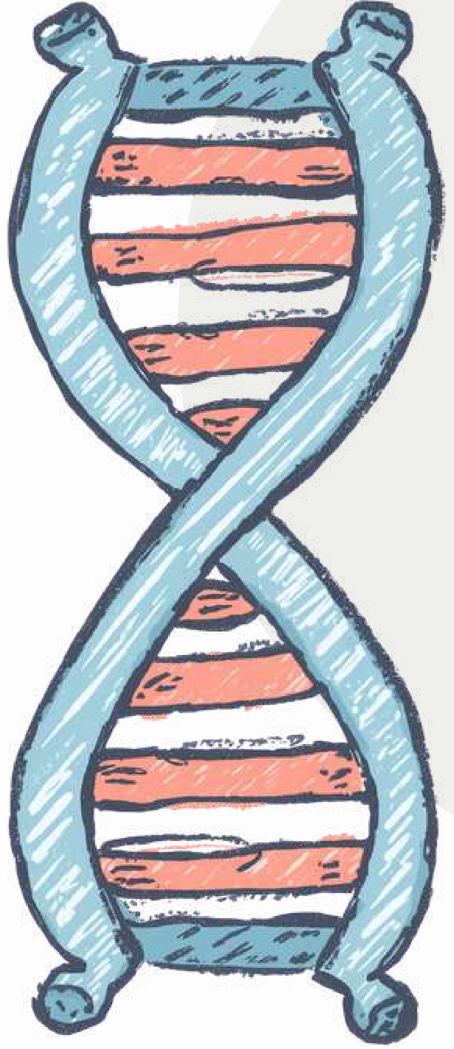
01 To study genes and traits that make us unique

02 To solve crimes using forensic science evidence

03 To improve crops and develop new medicines for people

04 To learn about living things and how they work

WHAT IS DNA EXTRACTION?



DNA extraction means separating DNA from the rest of the cell parts.

This process helps scientists see and study DNA clearly. By removing DNA from cells, we can examine the tiny instructions that make every living thing unique!

MATERIALS NEEDED FOR DNA EXTRACTION

01

Soap or detergent to break open cell walls

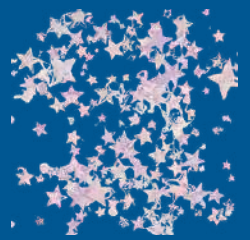
02

Salt to help clump the DNA together

03

Water and fruit like strawberries or bananas

MATERIALS NEEDED FOR DNA EXTRACTION (PART 2)



04

Coffee filter or cloth to strain the mixture

05

Chilled rubbing alcohol or ethanol

06

Clear container or test tube to collect DNA

STEP 1 — PREPARE THE FRUIT SAMPLE

Mash the fruit gently to break open the cells. This releases the DNA from inside! Use a fork or your hands to squish strawberries or bananas in a bowl until smooth.

STEP 2 — ADD SOAP SOLUTION

Mix soap, salt, and water together to create a special solution. This mixture helps dissolve the cell membranes and release the DNA from inside the fruit cells.

STEP 3 — FILTER THE MIXTURE

Pour the mixture through a coffee filter or cloth to remove the solid bits. This leaves you with a clear liquid that contains the DNA!

STEP 4 — ADD COLD ALCOHOL

Slowly pour cold alcohol on top of your filtered mixture. The alcohol makes the DNA come out of the liquid as white, stringy threads that you can see!

STEP 5 — OBSERVE THE DNA

White, stringy DNA will appear between the layers! Use a stick or glass rod to gently lift the DNA strands out. You can now see the tiny instructions that make living things work!

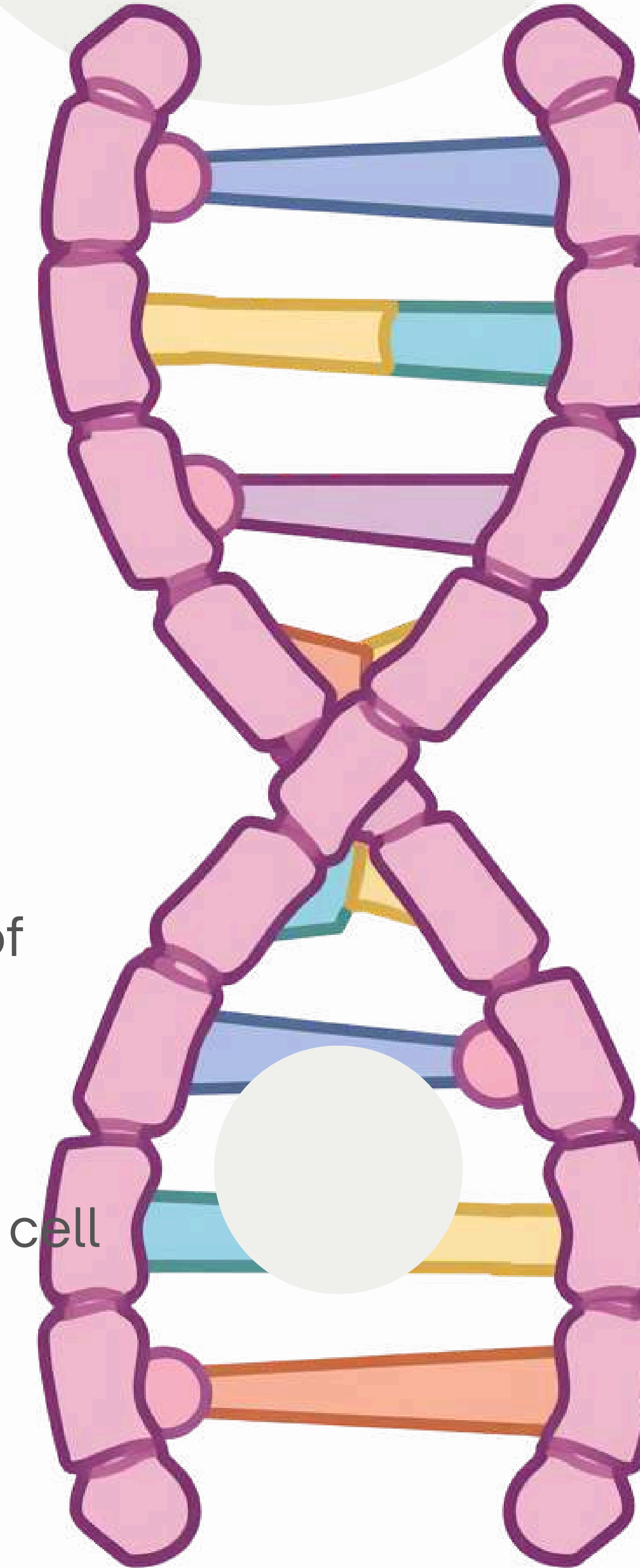
FUN FACTS ABOUT DNA

Amazing things you should know about DNA:

01 DNA is shaped like a twisted ladder called a double helix

02 Humans share about 99.9% of DNA with each other

03 DNA is found in almost every cell of your body

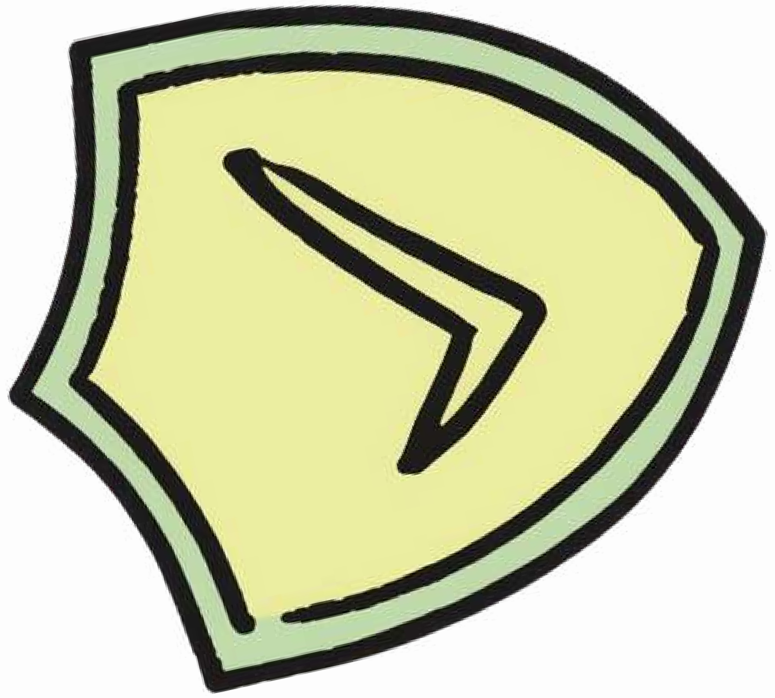


REAL-LIFE APPLICATIONS OF DNA EXTRACTION

DNA extraction helps scientists and experts solve important problems in many fields.

Solving crimes through forensics • Helping farmers grow better and stronger crops • Developing new medicines to cure diseases • Understanding genetic conditions

REMEMBER THESE SAFETY TIPS



Follow these important guidelines to stay safe during your DNA extraction experiment:

01 Always ask an adult before handling chemicals

02 Use clean equipment for accurate results

03 Wash hands thoroughly after the experiment

04 Work carefully and slowly throughout the process

SUMMARY — WHAT WE LEARNED TODAY

Key takeaways from our DNA extraction journey:

- 16 Biotechnology helps us understand and use living things
- 16 DNA is the instruction book inside every cell
- 16 DNA extraction uses simple materials like soap and alcohol

THANK YOU!

Thank you for exploring DNA with us! Try the experiment safely at home or school, and keep asking questions!